SQL PROJECT ON PIZZA SALES





ABOUT THIS PROJECT



Hi, myself Mayank Sharma and i am working on The Pizza Sales Analysis project aims to analyze pizza sales data using **SQL** to understand **sales performance,** customer preferences, and business trends. This project focuses on identifying the best-sellingpizzas, popular pizza sizes, top-performing categories, total revenue, and sales trends over time. By using SQL queries, the project transforms raw sales data into meaningful insights that can help businesses improve sales strategies, understand customer behavior, and make data-driven decisions.

THERE ARE SOME QUERIES I WORKED ON:-

- Basic:
- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Identify the highest-priced pizza.
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.

- Intermediate:
- Join the necessary tables to find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.

- Advanced:
- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

• RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED



```sql
SELECT
    COUNT(order_id) AS total_orders
FROM
    orders;
```

	total_orders
▶	21350

• CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES



```
SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS toal_revenue
FROM
    order_details
        JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id;
```

Result Grid

	toal_revenue
▶	817860.05

• IDENTIFY THE HIGHEST-PRICED PIZZA



```sql
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid | Filter Rows

name	price
The Greek Pizza	35.95

• IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.



```sql
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS count_order
FROM
    pizzas
        JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY COUNT(order_details.order_details_id) DESC;
```

Result Grid | Filte

size	count_order
L	18526
M	15385
S	14137
XL	544
XXL	28

• LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.



```sql
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows:

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

• JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.



```sql
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Result Grid | Filter Ro

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

• DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.



```sql
SELECT
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

Result Grid | Filter Rows:

hour	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920

- **JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.**



```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Result Grid | Filter Rows:

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

• GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY



```sql
SELECT
    ROUND(AVG(quantity), 0) as avg_quantity
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid

	avg_quantity
▶	138

• DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.



```sql
select pizza_types.name, sum(order_details.quantity * pizzas.price) as revenue
from pizza_types join
pizzas on
pizza_types.pizza_type_id= pizzas.pizza_type_id
join order_details on
order_details.pizza_id= pizzas.pizza_id
group by pizza_types.name
order by revenue desc limit 3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

• CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.



```sql
select pizza_types.category,
round(sum(order_details.quantity * pizzas.price) / (select round(sum(order_details.quantity * pizzas.price), 2) as total_sales
from order_details join
pizzas on
order_details.pizza_id= pizzas.pizza_id) *100,2) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id=pizzas.pizza_type_id
join order_details on
order_details.pizza_id= pizzas.pizza_id
group by pizza_types.category
order by revenue desc;
```

Result Grid | Filter Rows:

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

• ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.



```
select order_date,
sum(revenue) over(order by order_date) as cum_revenue
from
(select orders.order_date,
sum(order_details.quantity* pizzas.price) as revenue
from order_details join pizzas
on order_details.pizza_id= pizzas.pizza_id
join orders
on orders.order_id=order_details.order_id
group by orders.order_date) as sales;
```

Result Grid | Filter Rows:

order_date	cum_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5

• DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.



```sql
select name, revenue from
(select category, name, revenue,
rank() over(partition by  category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum((order_details.quantity) * pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id= pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <= 3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25

THANK YOU

